

SUBHRANSU PRIYARANJAN NAYAK

+91-7846860041 | +91-9937944664 | subhransu.nayak.connect@gmail.com | 21me02007@iitbbs.ac.in
github.com/NayakSubhransu | linkedin.com/in/subhransu-p-nayak | [LeetCode: Subhransu_Nayak_07](#)

Education

Indian Institute of Technology (IIT) Bhubaneswar

2021 – 2026

Integrated B.Tech in Mechanical System Design Engineering

CGPA: 8.3 | Odisha, India

Achievements

- **Competitive Programming & DSA:** Solved over **1,500+** problems | [LeetCode](#): Peak Rating **1934** (Top 4%, **1000+** Problems solved) | [CodeChef](#): Peak Rating **1682 (3-Star)** | [Codeforces](#): Peak Rating **1345 (Pupil)** (**200+** Problems solved).
- **Bronze Medalist: Inter IIT Tech Meet 10.0 (2022):** Secured **3rd place** for IIT Bhubaneswar across **23 IITs** [\[Certificate\]](#).
- **Intra-IIT Bhubaneswar Hackathon: 3rd place** in Game Dev (UE5) [\[Code\]](#) | **3rd place** in WebDev (ERP System) [\[Code\]](#).
- **IIT Bhubaneswar's Official Mobile App:** Contributed to the development of a **cross-platform mobile app** [\[Code\]](#) for GC 2024.

Experience

Data Engineering Intern | [HealthyDay \[Offer Letter\]](#) Hybrid, India | Nov 2025 – Present (3Mo. + 6Mo. Extended)

- Engineered a **BigQuery**-based data processing layer for a platform serving **600K+ registered students**, integrating Firestore and relational operational data by developing **50+ reusable SQL views** with CTEs, window functions, JSON parsing, regex classification, and temporal deduplication.
- Designed and optimized **150+ complex SQL queries** and data pipelines across user **attendance, referral, subscription, payment, and messaging workflows** using incremental aggregation strategies and query tuning.
- Built **end-to-end data pipelines** unifying heterogeneous sources (**Firestore, Google/Meta Ads, WhatsApp/AiSensy messaging logs, transactional databases**), implementing entity matching, and event sequencing across system boundaries.
- Developed automated data models and analytical views powering **Looker Studio** dashboards, implementing **funnel/event processing, cohort retention analysis, referral tracking, customer support metrics, and AI-classifier log monitoring**.

Software/AI Engineering Intern | [Hanyaa Auto Technologies \[Experience Letter\]](#) Hyderabad, India | Jun 2025 – Jul 2025

- Engineered an **AI text-to-animated-video narration pipeline** for Indian languages (EN, HI, TE) by fine-tuning **Dia-TTS (1.6B)**, optimizing training with dynamic LR scheduling and gradient clipping to eliminate robotic artifacts.
- Architected a long-form audio generation engine with sentence chunking (**NLTK**), input text normalization, failure-handling retry logic, and seed-locked speaker consistency (**IS11 tag**) for coherent synthesis.
- Built an interactive **Streamlit UI** integrating **Gemini 2.5 Flash TTS API** alongside **proprietary media pipelines**(Hanyaa's), enabling automated multilingual narration generation.

Projects

DynamoCore [\[Code\]](#) | A Highly Available Key-Value Storage| **Java, RocksDB, Maven, Gossip Protocol, Vector Clock**

- Implemented the [Amazon Dynamo paper \(2007\)](#) end-to-end using Core Java and its multithreading features, built a distributed **AP key-value store** with asynchronous quorum consensus ($N=3, W=2, R=2$) via non-blocking **CompletableFuture** pipelines, achieving **12 ms p99 write latency** under concurrent load.
- Implemented a **consistent hashing ring**(150 virtual nodes/host, MurmurHash3) with $O(\log N)$ key routing, achieved **sub-4% load distribution deviation** and bounded node-join/leave data migration to K/N , eliminating full-ring reshuffling on topology changes.
- Sustained **100% Writes** during primary replica failures using **sloppy quorums**, hinted handoffs, and a **P2P Gossip failure detector** (ALIVE/SUSPECT/DEAD) with automated background recovery replay, zero writes lost during simulated node outages.
- Eliminated the full-dataset reconciliation scans by implementing **Merkle Tree anti-entropy** with $O(1)$ root-hash validation (**6 ms p99**) – used immutable **Vector Clocks** for causal conflict detection, enabling targeted replica repair without global consistency locks.

UrbanEats [\[Code\]](#) | Event-Driven Food Delivery Platform | **React, Node, TypeScript, RabbitMQ, Socket.io, Docker**

- Decoupled core domains into **6 autonomous Node.js/TypeScript microservices** communicating via HTTP and asynchronous **RabbitMQ AMQP message queues**, ensuring fault tolerance and zero-downtime service autonomy.
- Built a real-time courier dispatch using **MongoDB 2dsphere spatial indexing** and **\$nearSphere** queries to match the nearest available courier within **500 m** – streamed live GPS telemetry at 10-second intervals to React/Leaflet clients via **Socket.IO** room-based routing.
- Automated abandoned cart with **MongoDB TTL indexes**(15-mins) and secured payment finalization with **Razorpay HMAC SHA-256** signature validation – used **RabbitMQ** to atomically persist confirmed transactions, preventing double-processing on retries.
- Optimized the media pipelines by streaming Multer in-memory buffers as **Base64 DataURIs to Cloudinary CDN** for zero-disk asset uploads, utilizing native MongoDB drivers to eliminate ODM overhead during bulk KYC verification workflows.

Enterprise RAG Orchestrator [\[Code\]](#) | **Agentic RAG & Text-to-SQL Engine** | **FastAPI, LangGraph, Qdrant, Redis**

- Built a **stateful multi-path reasoning engine** with **LangGraph** and PostgreSQL checkpoints – dynamically routes queries across RAG, Text-to-SQL, and Hybrid paths; implemented **Corrective RAG (CRAG)** web fallbacks and **Self-RAG** reflection loops to reduce hallucinations across a **95% noise document corpus**.
- Engineered a multi-stage retrieval pipeline with **Qdrant hybrid search (Dense + Sparse TF-IDF via RRF)**, **HyDE expansion**, and **Cross-Encoder reranking (Top-20 to Top-5)**, backed by a 5-tier cryptographic **Redis cache**.
- Developed an **AST-guarded Text-to-SQL engine** over 7 PostgreSQL 16 tables deployed on K8s – enforced **SELECT-only** execution policies and integrated **LangGraph interrupt()** **HITL** approval gates to block AI-generated mutation queries before execution.
- Hardened a **9-layer defense architecture** across FastAPI endpoints with ML prompt injection scanning (**11m-guard**), **XML spotlighting**, **bidirectional PII redaction**, token budgeting, and Ragas benchmarking.

Additional Projects: [transaction-ledger-backend](#) (*Idempotency Keys, Atomic Transactions, Ledger Model, JWT Blacklisting*) | [realtime-collaborative-editor](#) (*Liveblocks, Convex, Zustand, ProseMirror, Tiptap, Clerk, Zod, TypeScript, Next.js, React Hook Form*)

Skills

- **Languages:** Java, Python, C, C++, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Core CS & System Design:** Data Structures & Algorithms, Operating Systems, Multithreading, Networking, DBMS, Distributed Systems, Microservices Architecture, High-Level Design, Design Patterns, OOPs, API Design, Data Modelling
- **Web & Frameworks:** React.js, Next.js, Node.js, Express.js, React Native, FastAPI, Streamlit, Socket.io, Tailwind, REST APIs
- **Databases & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, BigQuery, Firebase, Docker, RabbitMQ, Git, Linux, CI/CD
- **AI/ML & GenAI:** PyTorch, TensorFlow, LangGraph, RAG, Hugging Face, Transformers, Scikit-learn, XGBoost, NLTK

Certifications

[System Design \(Grokking\)](#) (2026), [System Design Basics](#) (2026), [Advanced SQL Querying](#) (2025), [SQL](#) (2023), [Algorithms & CP](#) (2025), [Java In-Depth](#) (2023), [Deep Learning](#) (2023), [Machine Learning](#) (2023), [The React.js Guide](#) (2025), [The React Native Guide](#) (2025)